From Accounts to Tokens: Investment and Risk in a Stablecoin Economy

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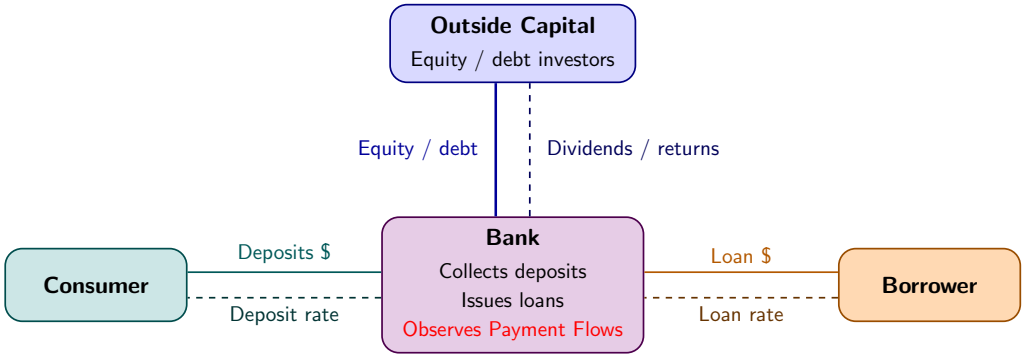
Introduction

- Current payment system is account-based through banks.
- Banks are also important investors in the economy.

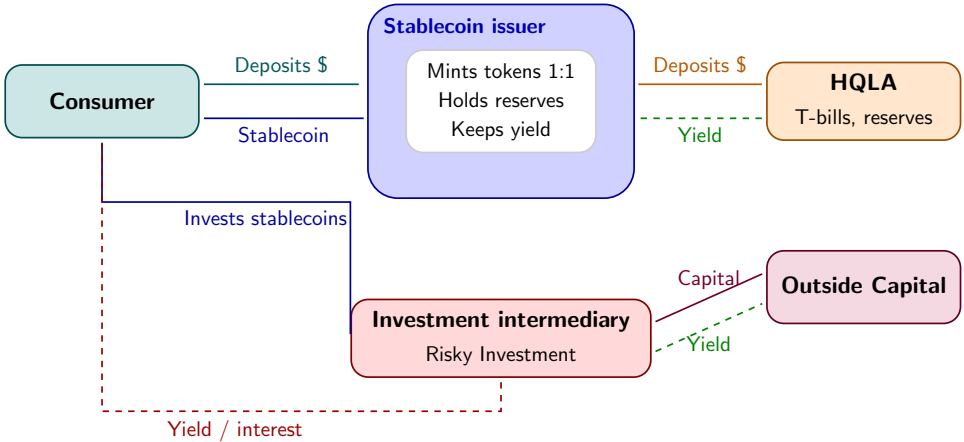
- Suppose we move to stablecoin payments (token-based).
- On-chain investment intermediaries will develop to fulfill the investment role of today's banks.

- Q: What implications and risks does this have for real projects in the economy?
  - We develop a stylized model to explore this question.
  - We show that left-tail risk increases in the economy.

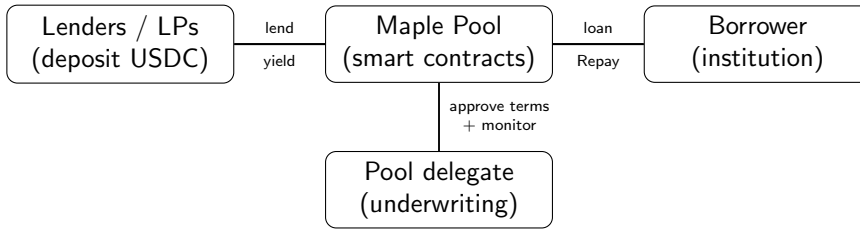
Bank Intermediated Payments



Stablecoin Intermediated Payments: No information Aggregation



Example: Maple Finance, $4.5 b. AUM



- Loans are often overcollateralized
- Borrowers vetted by delegates (off Chain)
- Lenders obtain fixed yield
- Also stakers provide insurance against losses

Account Based vs Token Based Payment system

- Account based payment system generates useful information about payment types
  - Maintained balances, payment prioritization, cashflow and liquidity management
- Useful to understand cashflows/repayment in bad states
- Not particularly informative about growth options or future plans
- Account based payments systems are better at evaluating default risk

Projects

- There are a continuum of firms. Each firm has a project that costs I at date 0. Firm needs to raise funding from external financiers.
- Project produces a random cash flow $x \in \{x_h, x_m, x_\ell\}$ at date 1. Set $x_\ell = 0$.
- Probabilities over states (p_h, p_m, p_ℓ) are initially unknown both to firm and financiers.
- Assume:
  1. All parties are risk-neutral, risk-free rate $= 0$.
  2. Dirichlet prior over states, with $\alpha_h = \alpha_m = \alpha_\ell = 1$. i.e., each point is equally likely.
  3. Average project is positive NPV: $\frac{x_h + x_m}{3} \geq I$.

Financier and Information

- Suppose financier holds a security that has realized payments

$$\begin{array}{cc} y_h & \text{in state } h \\ y_m \leq y_h & \text{in state } m \\ 0 & \text{in state } \ell \end{array}$$

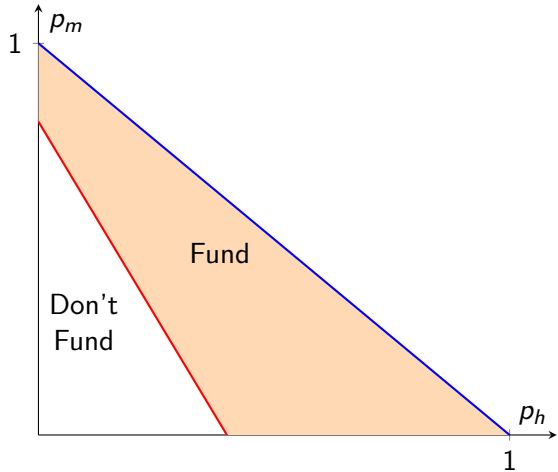
  Here, $y_h \in [0, x_h]$ and $y_m \in [0, x_m]$.

- E.g., Debt may have payments $D, D, 0$ if $x_m \geq D$.
  Equity has payments $\gamma x_h, \gamma x_m, 0$.

- Assume $\frac{y_h + y_m}{3} \geq I$: With no further information, project is funded.
  Denote $V_0(y_h, y_m) \equiv \frac{y_h + y_m}{3} - I$.

- Suppose financier can discover one of the three probabilities, p_h, p_m, or p_ℓ.
  Which one would they want to discover?

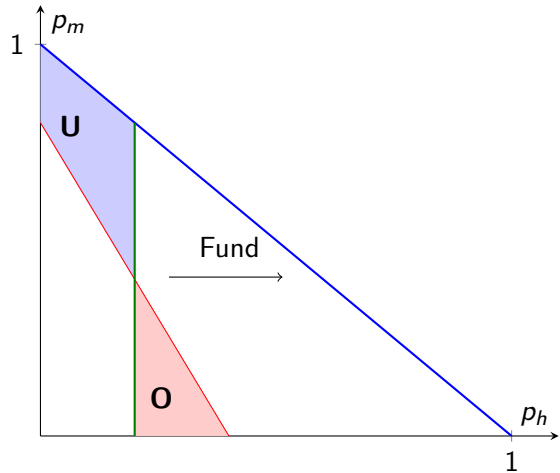
Benchmark: First-Best Investment



- Projects lie in triangle defined by axes and $p_h + p_m = 1$. Note that $p_\ell = 1 - (p_h + p_m)$.
- First-best investment specifies funding iff

$$p_h x_h + p_m x_m \geq I \qquad (1)$$

Funding Decision if p_h is Discovered



- If p_h is known, project is funded only if

$$p_h y_h + \frac{1 - p_h}{2} y_m \geq I$$

$$p_h \geq \frac{2I - y_m}{2y_h - y_m} \equiv \underline{p}_h$$

- Three potential frictions relative to first-best:
  1. Incomplete information.
  2. Uncompetitive financing.
  3. Cost of information.

Value of Discovering p_h

- Density over p_h is $2(1-p_h)$ over $[0,1]$.
- Project value conditional on knowing p_h:

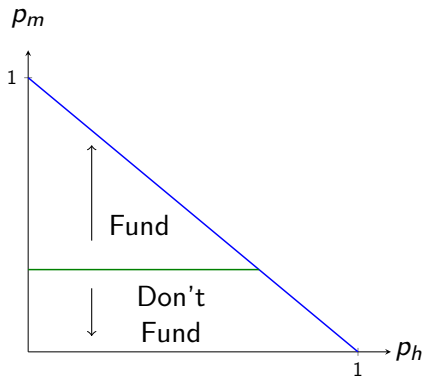
$$V_h = 2\int_{\underline{p}_h}^{1}\left\{py_h + \frac{1-p}{2}y_m - I\right\}(1-p)dp \tag{2}$$

$$= \frac{4(y_h-I)^3}{3(2y_h-y_m)^2} \quad \text{if } y_m < 2I. \tag{3}$$

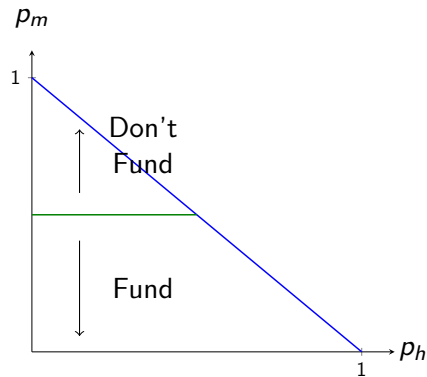
- Value of discovering $p_h = V_h - V_0$:

$$\Delta_h = \begin{cases} \left(\frac{2I-y_m}{2y_h-y_m}\right)^2 (y_h - \frac{y_m+I}{3}) & \text{if } y_m < 2I \\ 0 & \text{if } y_m \geq 2I. \end{cases} \tag{4}$$

Funding Decision if p_m is Discovered



$y_m > \frac{y_h}{2}$



$y_m < \frac{y_h}{2}$

Value of Discovering p_m

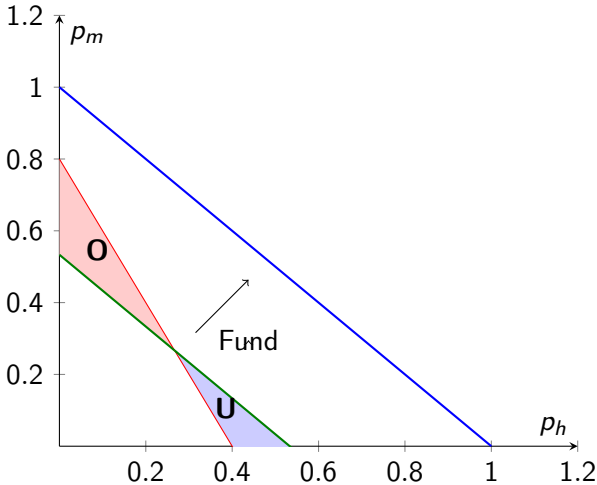
- If $y_m > \frac{y_h}{2}$ ($y_m < \frac{y_h}{2}$) a high value of p_m is good (bad) news.
- Project value conditional on knowing p_m depends on the case. For $y_h < 2\max\{I, y_m\}$

$$V_m = 2\int_{\underline{p}_m}^{1} \{\frac{1-p}{2} y_h + p y_h - I\}(1-p)dp. \tag{5}$$

- Value of discovering p_m:

$$\Delta_m = \begin{cases} \frac{(2I-y_h)^2(3y_m-y_h-I)}{3(2y_m-y_h)^2} & \text{if } y_h < 2\max\{I, y_m\} \\ \frac{4(I-y_m)^3}{3(y_h-2y_m)^2} & \text{if } y_m < \max\{I, \frac{y_h}{2}\} \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

Funding Decision if p_ℓ is Discovered



- Fund project if

$$\frac{1-p_\ell}{2}(y_h+y_m) \geq I$$

$$\implies p_\ell \leq \bar{p}_\ell \equiv 1 - \frac{2I}{y_h+y_m}$$

- Again there are over-funding and under-funding regions relative to first-best.

Value of Discovering p_ℓ

- Project value conditional on knowing p_ℓ:

$$V_\ell = 2\int_0^{\bar{p}_\ell} \{py_h + \frac{1-p}{2}y_m - I\}(1-p)dp \tag{7}$$

$$= \frac{y_h + y_m}{3} - I + \frac{4I^3}{3(y_h + y_m)^2}, \tag{8}$$

- Value of discovering p_ℓ:

$$\Delta_\ell = \frac{4I^3}{3(y_h + y_m)^2}. \tag{9}$$

Which Probability Does the Financier Want to Know About?

MICHIGAN ROSS

Proposition

Suppose a financier holds a security (y_h, y_m), where $y_h \geq y_m$ and $y_s \in [0, x_s]$ for each $s = h, m$. Further, suppose the financier can discover one of the probabilities p_h, p_m, or p_ℓ at no cost. Then:

(i) *The financier strictly prefers to discover p_h if either (a) $y_m \leq I$ or (b) $y_m \in (I, 2I)$ and $y_h > z(y_m)$, where $z(y_m) > y_m$ is a strictly increasing function.*

(ii) *For every other security, the financier prefers to discover p_ℓ.*

- If $y_m < 2I$ and the gap between y_h and y_m is large, optimal to discover p_h.
- Else optimal to discover p_ℓ.
- Never optimal to discover p_m.

Debt vs Equity

- Consider debt with realized payments (D, D) in states (h, m).
  - Feasible if $y_m \geq \frac{3}{2}I$.
  - Makes it optimal to discover p_ℓ.
- Consider equity with realized payments $(\gamma x_h, \gamma x_m)$. If (i) x_m is low or (ii) γ is low and x_h is high, optimal to discover p_h.

Banks and VCs

- Banks provide payment services for firms through an account-based system.
  - As a by-product, discover p_ℓ for free. Large literature, including Puri, Rocholl, Steffen (2017).
  - Also, banks make loans to firms.
- VCs take bets on high outcomes.
  - Due diligence to uncover p_h.
  - VCs hold equity-like claims.
- In each case, information set and financial claim are matched to each other.

Bank Discovering p_h?

- Suppose a bank knows p_ℓ. What is the incremental value of knowing p_h?
  - Knowing p_h as well leads to full information.

- $\Delta_{h|p_\ell} = V_f - \frac{(y_h+y_m-2I)^2(y_h+y_m+I)}{3(y_h+y_m)^2}$.

Proposition

There exists a strictly decreasing function $\psi(y_h)$ such that $\Delta_h > \Delta_{h|p_\ell}$ if and only if $y_m > \psi(y_h)$.

- In particular, for a debt security (D, D) there is zero incremental value to knowing p_h given p_ℓ.

Bank Payments vs Stablecoin Payments

- Banks provide account-based payments.
  - Discover p_ℓ in the process.
  - Invest in projects when p_ℓ is low enough.
  - Eliminate some left-tail risk.

- Stablecoins are a token-based payment system.
  - On-chain intermediaries will need to funnel consumer savings to firms.
  - These intermediaries will have to choose whether to find information about high states or low states.
  - Under many conditions, will focus on high states.
  - Over-invest in projects with intermediate probabilities of state h.

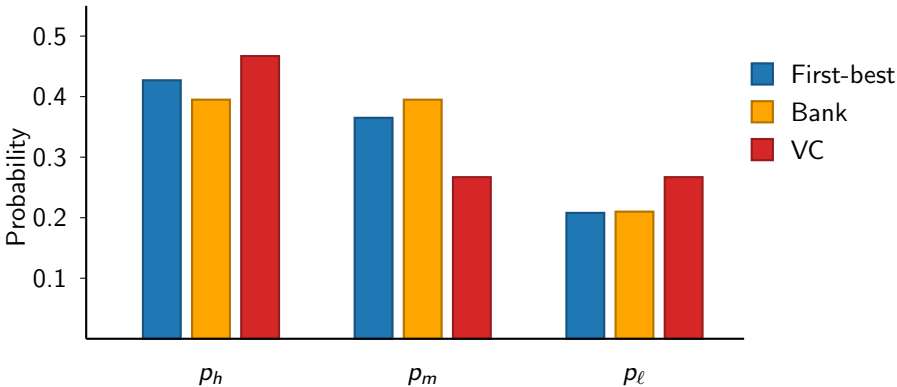
- Transition from bank payments to stablecoin payments affects risk of real projects in economy.

Aggregate Outcomes

- Suppose financing is competitive, and mass of projects is 1.
- Numerical example comparing Bank vs VC financing. Set $x_h = 2, x_m = 1, I = 0.8$.

	Mass of Projects Invested in	Aggregate Value
First-best Investment	0.680	0.285
Bank Financing	0.716	0.276
VC Financing	0.640	0.256

Probability Distributions Across Financing Regimes



Implication 1: Relationship to Tokenized Deposits

- Tokenized deposits are fundamentally different to stablecoins because they are account-based.
- Tokenization provides extra layer of functionality.
- Can have similar regulatory structure to banks.

Implication 2: Value of Information

- Is it possible to replicate information about p_ℓ through another mechanism?
- No: wallet to wallet transfer requires both sides to report
- Clear need for regulatory solution to control risk-taking through stablecoin investments.

Implication 3: Liquidity Transformation

- Financial intermediaries choose illiquid investments
  - If funding comes from payments demand $\longrightarrow$ mismatch
- DeFi solution: issue tradable tokens ("staked").
  - Tokens are claims to future returns. Jacklin (1987).
- Ameliorates mismatch but singleness of money problem.
- Clear regulatory concern: Liquidity in tokens market may dry up.

Final Thoughts

- Account-based banking systems are economically different to token-based systems
- Macro implications: Set of projects undertaken without bank-like information become more extreme.
- Emerging on-chain intermediation structure requires new regulatory approaches.